

TITLE PAGE

- **Problem Statement Title: Smart Education**
- **Theme: Smart Education**
- **PS Category: Software**
- **Team Name: Heisyn**
- **Project Name: Codify**



Proposed Solution: *Codify: Empowering tomorrow's professionals, today.*

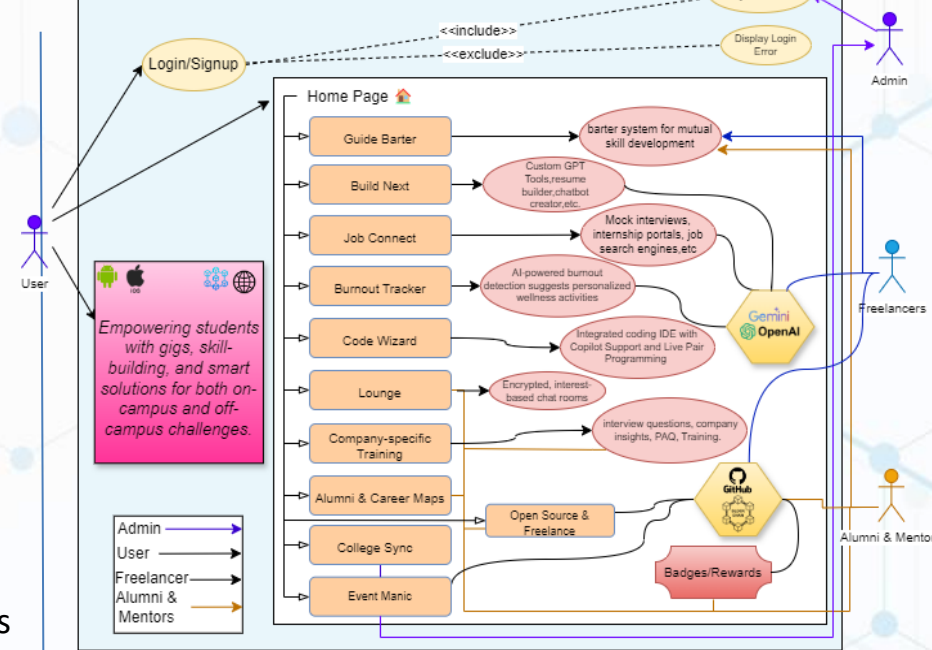
Codify provides a unified and **all-in-one** platform to guide students and professionals by offering essential **academic and professional tools**. It helps you **move seamlessly** from college to career with resources, connections, and **innovative features**.

-  **1. Guide Barter:** A **peer-to-peer** platform using smart contracts for skill exchange and guided mentorship.
-  **2. Build Next:** AI-driven **NLP tools** for resume generation, resume analyzer, and custom GPT model.
-  **3. Lounge:** **Encrypted**, interest-based chat rooms powered by **communication protocols**.
-  **4. Job Connect:** AI-powered **job-matching algorithms** with automated application submissions and startup-friendly portals.
-  **5. Inter(company's)view:** Machine learning-based mock interviews with **company-specific datasets and real-time feedback**.
-  **6. College Sync:** **Cloud-based** access to semester-wise PYQs, notes, and video tutorials with seamless synchronization.
-  **7. Alumni & Career Maps:** Blockchain-verified alumni networks and **AI-generated career road maps**.
-  **8. Code Wizard:** Integrated **coding IDE** and competitive coding platforms
-  **9. Open Source & Freelance:** GitHub **API integration** for tracking **open-source contributions** and freelance work, earning **badges**.

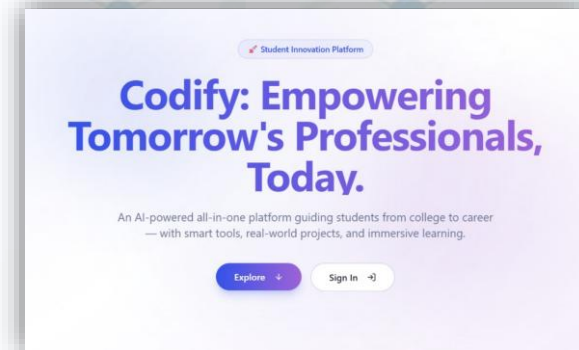
Our Stand Out Feature:

AI free Interview Hub: Codify offers immersive AI based audio interview covering every type of roles and interview complexity.

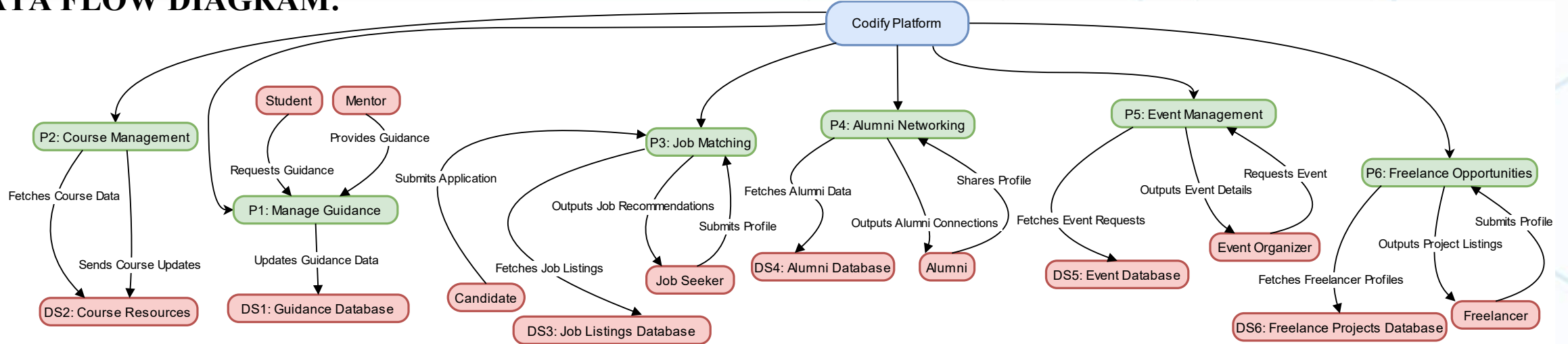
USE CASE DIAGRAM:



DEMO of our web UI/UX...



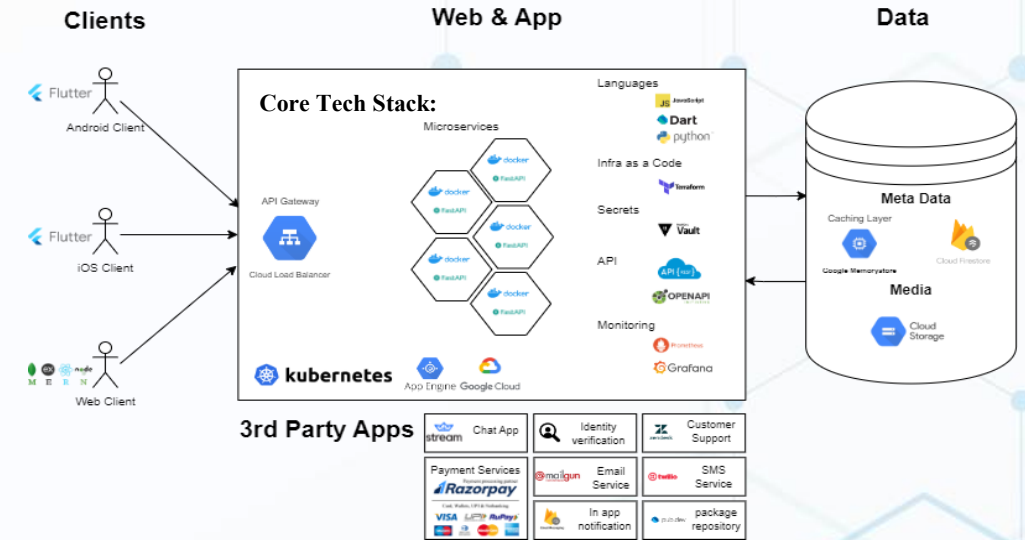
DATA FLOW DIAGRAM:



TECH STACK:

- **Tech Stack & Dependencies:**
- **PERN Stack (PosgresDB, Express, React, Node.js)** with **Next.js** for server-side rendering.
- **Express Js** with RESTFULL APIs for state management and **Node Js** for real-time data, authentication, and **nginx** for VPS server hosting.

- **AI Technologies:** AI LLMs like Gemini api
- **Version Control:** Git and GitHub.
- **REST APIs** for data exchange.
- **Libraries/Dependencies:**
- **Axios** or **Fetch** for API requests.
- **Redux** or **Context API** for React state management.



GITHUB REPO: <https://github.com/Shreya71703/HeisynYUKTI>

FEASIBILITY AND VIABILITY

USE CASES/ FEASIBILITY:

- **Proven Technologies:** Leverages React for responsive front-end development, AI-driven Natural Language Processing (NLP) for automation (resume building, chatbot creation, etc.), and Blockchain for decentralized verification (certifications, alumni networks).
- **Cloud Scalability:** Uses cloud-based architecture to ensure high performance, reliability, and real-time synchronization of resources, from study materials to interview simulations.
- **Smart Contracts & AI Integration:** Seamlessly integrates smart contracts for secure peer-to-peer skill bartering and AI algorithms for real-time job matching and personalized feedback.

SHOWSTOPPER/ CHALLENGES:

- **Resource Management:** Scaling resource libraries (notes, past year question papers, projects) across multiple educational institutions.
- **Sustained User Engagement:** Ensuring users remain engaged across multiple offerings like skill exchanges, mentorship, and job opportunities.

SOLUTIONS:

- **AI-Powered User Retention:** Implement personalized AI-driven notifications and learning pathways based on user behavior and interests.
- **Decentralized Incentives:** Introduce blockchain-verified rewards and badges for open-source contributions, skill acquisition, and freelance work to foster long-term engagement.
- **Cross-Platform Accessibility:** Ensure seamless access across devices (web and mobile), making resources and opportunities available anytime, anywhere.

VIDEO SOLUTION LINK: <https://drive.google.com/drive/folders/1o4sdqj6tS12KV9LkILfyJVV4wC1SmdT>

IMPACT AND BENEFITS

IMPACT:

Impact on Target Audience:

- **For Students & Professionals:** Access to industry-specific resources, AI-powered job matching, and skill development—all under one platform.
- **For Freelancers:** Build a recognized portfolio with verifiable contributions to open-source projects, making your work easily visible to recruiters.
- **For Universities & Companies:** Seamless integration of career pathways with verified alumni networks and decentralized mentorship programs.

Broad Impact:

- **Social Impact:** Fosters collaboration and continuous learning through decentralized peer-to-peer exchanges and mentorship.
- **Economic Impact:** Bridges the skills gap by connecting fresh graduates and professionals with internships and job opportunities.
- **Career Impact:** Provides personalized career paths and real-time feedback, enhancing job readiness and long-term professional growth.

BENEFITS:

Efficiency: Simplifies the journey from learning to employment using AI, Blockchain, and cloud infrastructure, minimizing barriers in job searching and skill verification.

Trust & Security: Blockchain ensures tamper-proof credentials, enhancing trust for recruiters and mentors.

Scalability: Cloud architecture supports large-scale resources and real-time synchronization.

User Retention: AI-driven incentives like personalized learning paths and skill badges keep users engaged long-term.

REVENUE STREAMS:

- **Subscriptions:** Charge for premium features like advanced tools, AI-driven services, and mock interviews.
- **Freemium:** Offer basic features free, with paid upgrades (e.g., custom GPT models, verified certifications).
- **Corporate Partnerships:** Companies pay for recruitment access to verified candidates.
- **Rental Collaborations Model:** Charge institutions for resource management and student career services.
- **Freelance Commissions:** Take a cut from freelance transactions on the platform.

RESEARCH AND REFERENCES

AI-Powered Mental Health Interventions for Young Adults:

<https://ai.jmir.org/2022/1/e38171>

Artificial Intelligence for Career Guidance– Current Requirements and Prospects for the Future:

<https://files.eric.ed.gov/fulltext/EJ1318705.pdf>

https://www.researchgate.net/publication/354171816_Artificial_Intelligence_for_Career_Guidance_-_Current_Requirements_and_Prospects_for_the_Future

Gamification in Education: What, How, Why Bother? by Educause Review:

https://www.researchgate.net/publication/258697764_Gamification_in_Education_What_How_Why_Bother

Peer Review and Feedback with AI: Enhancing Student Learning by MIT News:

<https://news.mit.edu/2016/using-educational-technology-enhance-student-learning-0805>

Gig Economy in Education: Opportunities and Challenges by World Economic Forum :

<https://www.weforum.org/agenda/2021/05/what-gig-economy-workers/>